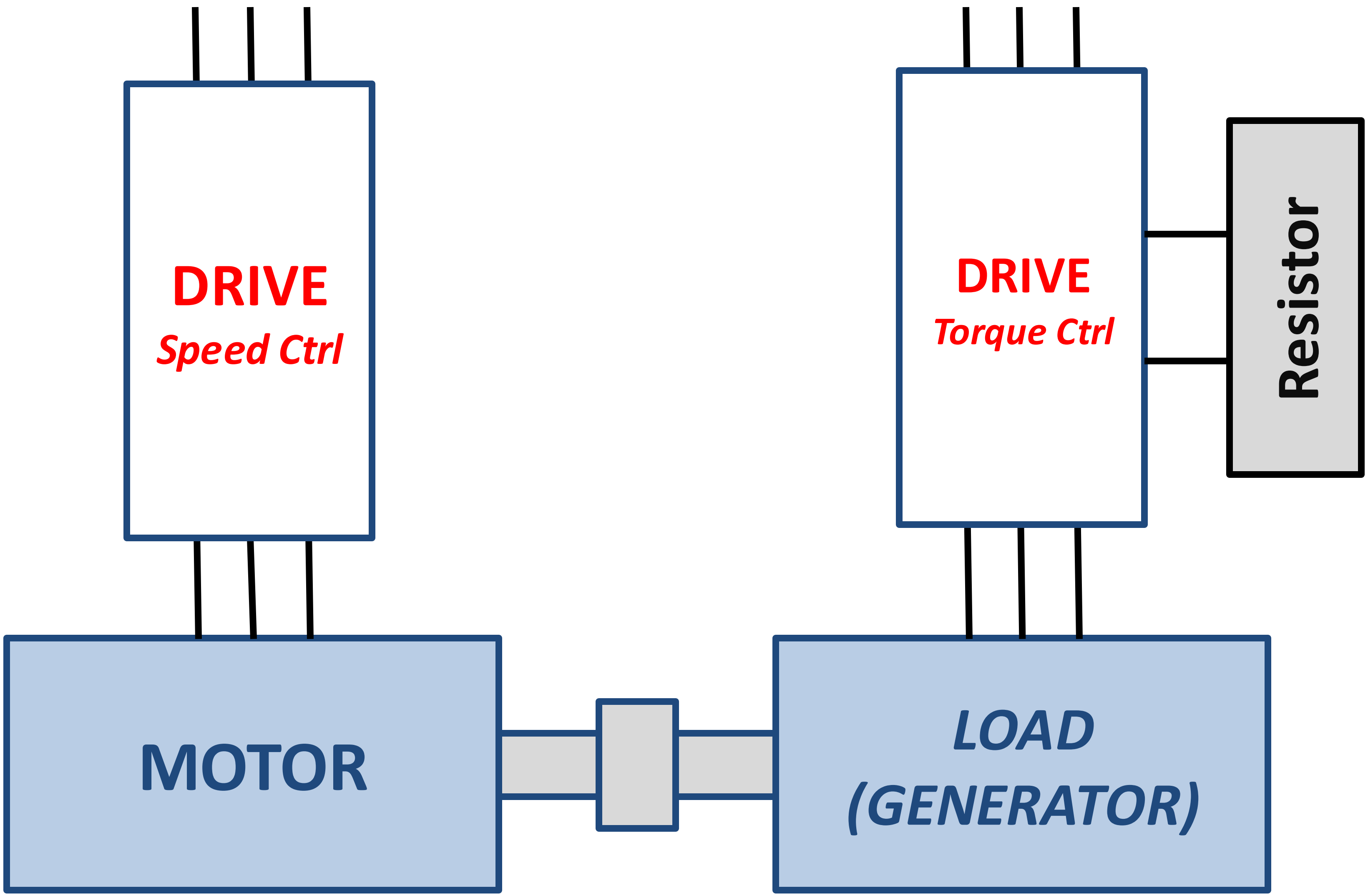
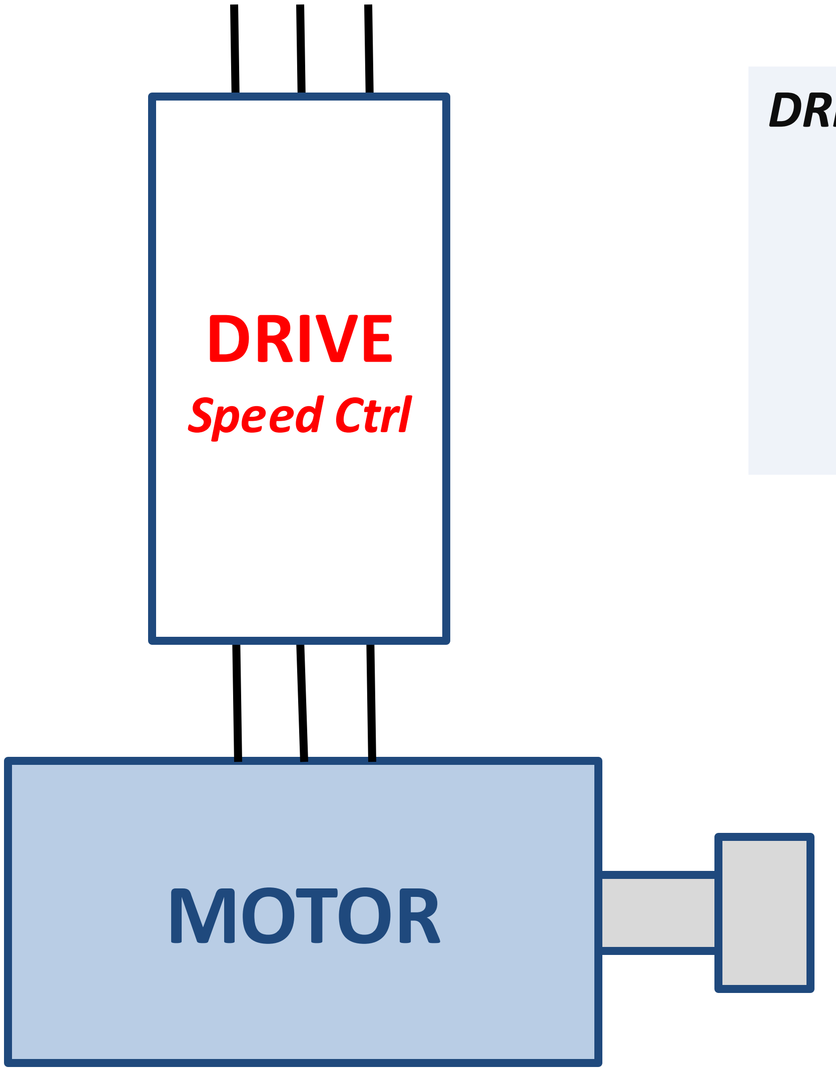
Test-Bed


DRIVE
Speed Ctrl
DRIVE
Torque Ctrl
Resistor
MOTOR
LOAD
(GENERATOR)

Test-Bed



DRIVES:

ACSM1-046A-4 C/ ACS850C

V: 400v

I_{IN}: 39A

I_{OUT}: 46A

P: 22kW

MOTOR / GENERATOR :

M4BP 160 MLB 3GBP

V: 400v

F=50Hz

I_N: 27.8A

T:97.1Nw/m

P: 15kW

1474rpm

Machines

- Healthy rotor
- Incipient broken rotor bar
- One broken rotor bar
- Multiple broken rotor bar

- Healthy bearing

- Groove Bearing

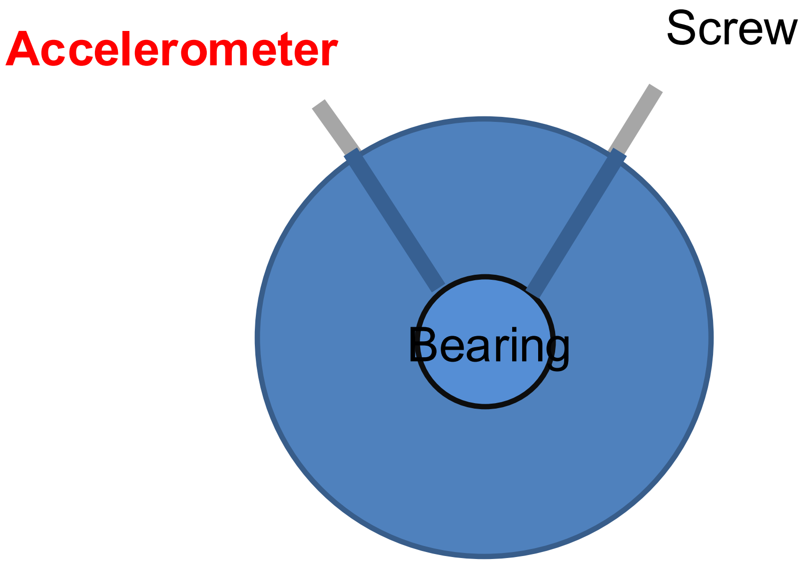
- Hole bearing (6mm)

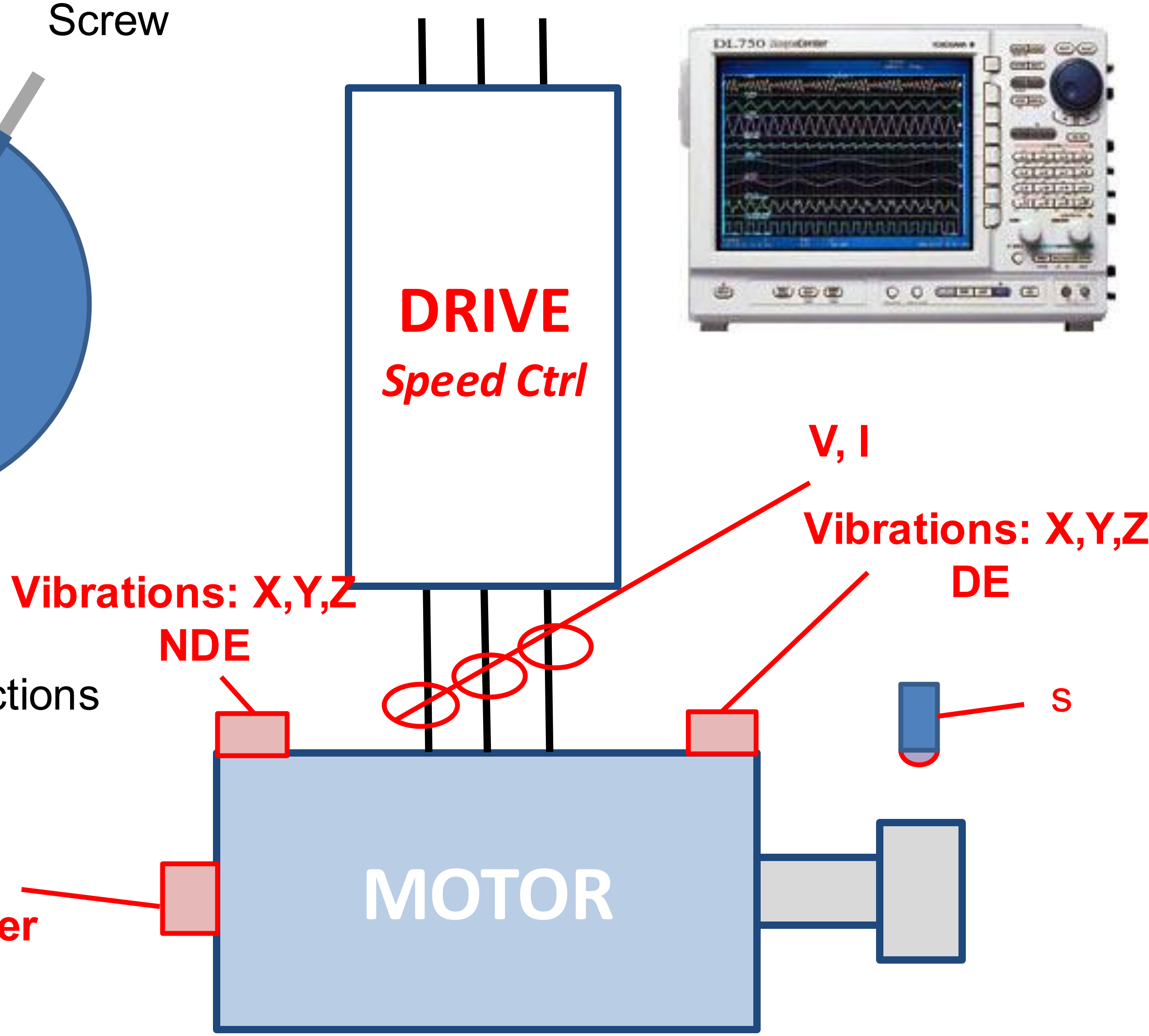
- Hole Bearing (3mm)

- Cool Bearing

- Stator Fault 2 (Max. Current)
- Stator Fault 3 (limited short-circuit current with a resistance of 20 ohm)

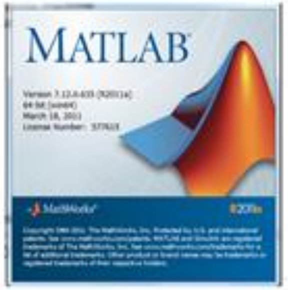
Measurements





- ACC measure in three directions
- Accelerometer only measure in one direction

Data Base Tests



1. Machines:
   - 'h' Healthy machine
   - 'e' Healthy machine coupled with One broken bar
   - 'b' One broken rotor bar
   - 'v' Several broken rotor bars
   - 'p' Prognosis broken rotor bar
   - 'g' Groove bearing damage
   - 'o' Hole 6mm bearing damaged
   - 'r' Hole 3mm bearing damaged
   - 'c' Cool bearing damaged (real)

2. Control:
   - 'l' Direct on-line Grid
   - 's' Scalar Control in Drive.
   - 'd' DTC Control in Drive.

3. Speed:
   1. 1500 r.p.m.
   2. 1200 r.p.m.
   3. 900 r.p.m.
   5. 20 r.p.m.
   6. 100-1500-100 r.p.m.
   7. 100-500-1000-1500-1000-500-100
   9. 0-1500 r.p.m. (start-up)

4. Load:
   0. 0 %
   1. 25 %
   2. 50 %
   3. 75 %
   4. 100 %
   6. 0-100-0-100-0 %
   7. 25-75-25-75-25 %
   8. 50-100-50-100-50 %

iu
iv
iw
a
s
AccNDy
AccNDz
AccNDEy
AccNDEz

Data Base Tests

1. **Machines:**
   - **'h' Healthy machine**
   - **'e' Healthy machine coupled with One broken bar**
   - **'b' One broken rotor bar**
   - **'v' Several broken rotor bars**
   - **'p' Prognosis broken rotor bar**
   - **'g' Groove bearing damage**
   - **'o' Hole 6mm bearing damaged**
   - **'r' Hole 3mm bearing damaged**
   - **'c' Cool bearing damaged (real)**

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   3. 900 r.p.m.
   5. 20 r.p.m.
   6. 100-1500-100 r.p.m.
   7. 100-500-1000-1500-1000-500-100
   9. 0-1500 r.p.m. (start-up)

4. **Load:**
   0. 0 %
   1. 25 %
   2. 50 %
   3. 75 %
   4. 100 %
   6. 0-100-0-100-0 %
   7. 25-75-25-75-25 %
   8. 50-100-50-100-50 %